

M3: Forces and Newton's Laws

1. Forces as Vectors

Forces are vector quantities — they have both magnitude and direction. The **resultant force** is the single force equivalent to all forces acting, found by vector addition.

Resultant Force: $\mathbf{F}_{net} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots + \mathbf{F}_n$

In component form: $F_x = \sum F_i \cos \theta_i$, $F_y = \sum F_i \sin \theta_i$

The magnitude of the resultant: $|\mathbf{F}_{net}| = \sqrt{F_x^2 + F_y^2}$, and its direction: $\theta = \arctan\left(\frac{F_y}{F_x}\right)$.

2. Newton's Laws of Motion

- **First Law:** A particle remains at rest or moves with constant velocity unless acted on by a resultant force.
- **Second Law:** $\mathbf{F}_{net} = m\mathbf{a}$ — the net force equals mass times acceleration (vector equation).
- **Third Law:** Every action has an equal and opposite reaction.

$$\mathbf{F}_{net} = m\mathbf{a}$$

For problems in a single direction, resolve forces along that direction and apply $F = ma$ as a scalar equation.

3. Common Forces in Mechanics

Force	Symbol	Definition
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Weight	W	$W = mg$ downward; $g = 9.8 \text{ m/s}^2$
Normal Reaction	R or N	Perpendicular to the surface, away from surface
Tension	T	Along a string/rod, pulling the particle
Friction	F	Along the surface, opposing motion or tendency to move
Thrust	P	Along a rod, pushing the particle (compressive)

4. Resolving Forces

Any force F at angle θ to the horizontal can be split into:

$$F_x = F \cos \theta \quad F_y = F \sin \theta$$

To solve problems: choose axes (usually horizontal/vertical or along/perpendicular to slope), resolve all forces, apply $F = ma$ in each direction.

Exam Tip: Always draw a clear force diagram before writing equations. Label every force with its magnitude and direction. Never omit the weight mg .

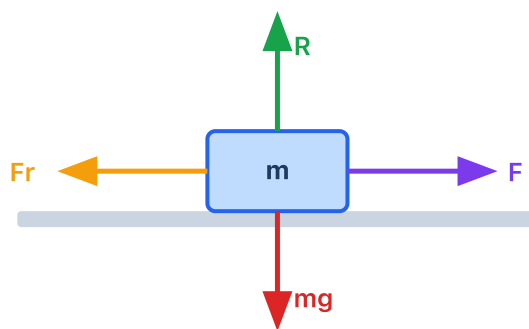
5. Equilibrium

A particle is in **equilibrium** when the resultant force is zero:

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

Equivalently, $\mathbf{F}_{net} = \mathbf{0}$, so $m\mathbf{a} = \mathbf{0}$ and the particle is at rest or moving at constant velocity.

Particle on a Horizontal Surface — Force Diagram



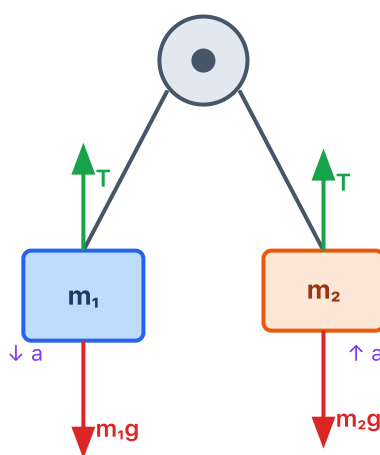
Particle on a rough horizontal surface: R (normal), mg (weight), F (applied), Fr (friction)

6. Connected Particles

When two particles are connected by a **light inextensible string** over a **smooth pulley** (Atwood machine):

- The string has the same tension T throughout (light string, smooth pulley).
- Both particles have the **same magnitude of acceleration** a (inextensible string).
- Write a separate $F = ma$ equation for each particle, then solve simultaneously.

Atwood Machine Setup



Atwood machine: $m_1 > m_2$, so m_1 descends with acceleration a

Atwood Machine Equations:

For m_1 (descending): $m_1g - T = m_1a$

For m_2 (ascending): $T - m_2g = m_2a$

Adding: $(m_1 - m_2)g = (m_1 + m_2)a$

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1m_2g}{m_1 + m_2}$$

Particle on a Table Connected to a Hanging Particle

Particle A (mass m_1) on a smooth horizontal table, connected by a string over a pulley at the edge to particle B (mass m_2) hanging freely.

- For A (horizontal, no friction if smooth): $T = m_1a$
- For B (vertical): $m_2g - T = m_2a$
- Solve: $a = \frac{m_2g}{m_1 + m_2}$, $T = \frac{m_1m_2g}{m_1 + m_2}$

7. Particle in a Lift

A particle of mass m stands on the floor of a lift. The normal reaction N (apparent weight) depends on the lift's acceleration a :

$$N = m(g + a) \quad (\text{accelerating upward})$$

$$N = m(g - a) \quad (\text{accelerating downward})$$

- If $a = 0$ (constant velocity): $N = mg$ (normal weight).
- If $a = g$ downward (free fall): $N = 0$ (weightlessness).

Exam Tip: In a lift problem, always define positive direction clearly (e.g., upward positive). Apply $F = ma$ in that direction. If lift accelerates upward, net upward force = $N - mg = ma$, giving $N = m(g + a)$.

8. Worked Example

Example: Two particles, A (3 kg) and B (5 kg), are connected by a light inextensible string over a smooth pulley. Find the acceleration and tension.

Take downward as positive for each particle in its direction of motion (B descends).

For B: $5(9.8) - T = 5a \Rightarrow 49 - T = 5a \dots(1)$

For A: $T - 3(9.8) = 3a \Rightarrow T - 29.4 = 3a \dots(2)$

Add (1)+(2): $19.6 = 8a \Rightarrow a = 2.45 \text{ m/s}^2$

Sub into (2): $T = 29.4 + 3(2.45) = 36.75 \text{ N}$

9. Key Formulae Summary

Formula	Use
$\mathbf{F} = m\mathbf{a}$	Newton's Second Law (vector)
$W = mg$	Weight ($g = 9.8 \text{ m/s}^2$)
$F_x = F \cos \theta, F_y = F \sin \theta$	Resolving components
$\sum F_x = 0, \sum F_y = 0$	Equilibrium conditions
$N = m(g \pm a)$	Apparent weight in lift
$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	Atwood machine acceleration

Common Errors to Avoid:

- Using $g = 10$ instead of $g = 9.8$ — Edexcel uses 9.8 unless told otherwise.
- Forgetting to write separate $F = ma$ equations for each particle in a connected particle problem.
- Wrong sign for tension direction — tension always pulls the particle toward the string/pulley.

- Forgetting that the string tension is the same throughout (light string, smooth pulley).